



Exploring the resistome and virulome in major sequence types of *Acinetobacter baumannii* genomes: Correlations with genome divergence and sequence types

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ABSTRACT

The increasing global prevalence of antimicrobial resistance in *Acinetobacter baumannii* has led to concerns regarding the effectiveness of infection treatment. Moreover, the critical role of virulence factor genes in *A. baumannii*'s pathogenesis and its propensity to cause severe disease is of particular importance. Comparative genomics, including multi-locus sequence typing (MLST), enhances our understanding of *A. baumannii* epidemiology. While there is substantial documentation on *A. baumannii*, a comprehensive study of the antibiotic-resistant mechanisms and the virulence factors contributing to pathogenesis, and their correlation with Sequence Types (STs) remains incompletely elucidated. In this study, we aim to explore the relationship between antimicrobial resistance genes, virulence factor genes, and STs using genomic data from 223 publicly available *A. baumannii* strains. The core phylogeny analysis revealed five predominant STs in *A. baumannii* genomes, linked to their geographical sources of isolation. Furthermore, the resistome and virulome of *A. baumannii* followed an evolutionary pattern consistent with their pan-genome evolution. Among the major STs, we observed significant variations in resistant genes against "aminoglycoside" and "sulphonamide" antibiotics, highlighting the role of genotypic variations in determining resistance profiles. Furthermore, the presence of virulence factor genes, particularly exotoxin and nutritional / metabolic factor genes, played a crucial role in distinguishing the major STs, suggesting a potential link between genetic makeup and pathogenicity. Understanding these associations can provide valuable insights into *A. baumannii*'s virulence potential and clinical outcomes, enabling the development of effective strategies to combat infections caused by this opportunistic pathogen.

1. Introduction

Acinetobacter baumannii is an aerobic Gram-negative opportunistic pathogen which is commonly found in immunocompromised individuals and in hospital-associated infections (Howard et al., 2012). It is one of the six most important multidrug resistant ESKAPE pathogens which include *Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter spp.* (Agyepong et al., 2023). However, the emergence and spread of multidrug-resistant (MDR) *Acinetobacter baumannii*, has not only been a major global health concern but described by the Infectious Disease Society of America as a global problematic nosocomial threat

also (Boucher et al., 2009; Woodford et al., 2011). Despite efforts to control contamination, *A. baumannii* remains a persistent cause of nosocomial infections, with emerging multi-drug-resistant (MDR) strains further exacerbating the clinical challenges. *A. baumannii*'s nosocomial impact is attributed to its MDR mechanisms, including alterations in target sites, efflux pumps, and outer membrane permeability (Verma et al., 2021). More specifically, *A. baumannii* is resistant to multiple classes of antibiotics including carbapenem, third-generation cephalosporins (Karah et al., 2012). The pathogen also expresses multiple virulence factors (VFs) like biofilm-associated proteins, lipopolysaccharides, and pili, aiding in antibiotic resistance and immune evasion (Aliramezani et al., 2019; Beceiro et al., 2013). VFs, which are primarily

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molecules which help pathogens to colonize the host at a cellular level (Sharma et al., 2017), help *A. baumannii* in exhibiting different mechanisms such as antibiotic resistance, environmental persistence, host-pathogen interactions and immune evasion. The rise of nosocomial *A. baumannii* infections and higher mortality rates associated with MDR strains highlight the urgency for effective surveillance and management (Nocera et al., 2021; Dijkshoorn et al., 2007; Sunenshine et al., 2007). Addressing antibiotic resistance in *A. baumannii* is critical for patient outcomes and public health. VFs play a key role in *A. baumannii*'s ability to cause disease and evade the host immune system (Sarshar et al., 2021). These factors, primarily associated with motility, adherence, and biofilm formation, contribute to *A. baumannii*'s survival and persistence under various environmental conditions.

Although routine phenotypic and molecular typing methods are valuable, Next-Generation Sequencing (NGS) offers an exceptional level of insight into the genomic structure of pathogens responsible for outbreaks. The utilization of next-generation sequencing (NGS) techniques combined with comparative genomics offers a promising avenue for tracking, monitoring, and ultimately controlling the spread of infections attributed to *A. baumannii*. By analysing the genetic makeup of various strains through NGS, researchers can discern similarities, differences, and evolutionary patterns, aiding in the identification of antibiotic resistance patterns, understanding VFs, and devising strategies for effective containment and management of *A. baumannii* infections. This comprehensive approach allows for a more precise understanding of the pathogen's behaviour and facilitates targeted measures to curb its spread within healthcare settings and communities. Sequencing the 16S or 23S of the pathogen as well as genomic methods like Multi-Locus Sequence Typing (MLST) or screening for genes like *rpoB*, *gyrB* or *OXA-type carbapenemase-encoding genes*, all have been successfully used as methodologies for tracking down infection and also controlling it (Shamsizadeh et al., 2017; Lee et al., 2017).

Given the clinical importance of *A. baumannii* infections, it is imperative to delve into genome diversity, epidemiology, transmission characteristics associated with antibiotic resistance mechanisms, and VFs. Our study aims to conduct a comprehensive comparative analysis of curated 223 *A. baumannii* genomes, decoding their genetic diversity concerning antibiotic resistance and VFs. MLST and pan genomic profiling revealed five major core clusters representing major STs in *A. baumannii* genomes. The findings of this study illuminate both the core and diverse aspects of antibacterial resistance and virulence mechanisms in this organism, emphasizing their association with genome dissemination. These insights offer potential research directions for developing therapeutic strategies to manage *A. baumannii* infections.

2. Materials and methods

2.1. Dataset retrieval and genome statistics

Genome-wide comparisons of all *A. baumannii* isolates irrespective of geographical locations were carried out. BV-BRC database (<http://www.bv-brc.org/>) (Olson et al., 2023) was screened for the complete genome sequences of *A. baumannii* isolates with the human host. Initially our search resulted in 496 sequences (as of June 2023). Further screening among these sequences were performed based on Genome Status, Genome Quality, *Homo sapiens* as Host Name, Sequencing Depth, presence of Assembly Accessions and GenBank Accessions. All sequences that were not completely annotated were also removed from the dataset. This led to selection of 223 *A. baumannii* sequences for the purposes of this study. The overall information about all the *A. baumannii* genomes which are part of the curated dataset, including the collection year, geographical location is provided in Supplementary Table 1.

2.2. Multi-Locus Sequence Typing (MLST) analysis

The web-based sequence typing of the 223 selected *A. baumannii*

genomes was conducted using the PubMLST database (<https://pubmlst.org/>; Jolley et al., 2018), which hosts the Oxford MLST scheme for *A. baumannii*. This scheme characterizes seven housekeeping genes: citrate synthase (*gltA*), DNA gyrase subunit B (*gyrB*), glucose dehydrogenase B (*gdhB*), homologous recombination factor (*recA*), 60-kDa chaperonin (*cpn60*), glucose-6-phosphate isomerase (*gpi*), and RNA polymerase $\sigma 70$ factor (*rpoD*). The alleles corresponding to these seven genes were identified from the 223 *A. baumannii* genomes using BLASTN against the PubMLST database with criteria of 90% identity and 90% alignment coverage as parameters. The alleles identified through BLASTN were then used to determine the allelic profiles and sequence types (STs) for each of the 223 *A. baumannii* genomes housed within the PubMLST database.

2.3. Pan-genome profiling

A pan-genome is a comprehensive representation of the entire gene content within a specific group of genomes (Willenbrock et al., 2007). It is composed of three main components: the core set, which includes genes present in all individual genomes of the group; the cloud set or the cloud genes, comprising genes unique to each genome of the group; and the shell set, which includes genes present in two or more, but not all, genomes of the group. In this study, we constructed the pan genome profile of the selected *A. baumannii* genomes using the high-speed pipeline Roary (Javkar et al., 2021; Page et al., 2015). Roary utilizes annotated assemblies of genomes generated from Prokka (Seemann, 2014) as its input and provides a pan genome profile for the input genomes, along with a core genome alignment file. Roary was implemented using BLASTP parameters of 95% identity and the rest were kept default.

2.4. Prediction of antibiotic resistance genes (ARGs)

To estimate the presence of ARGs corresponding to various resistance mechanisms in the *A. baumannii* genomes, an *in-silico* analysis was performed using the Resistance Gene Identifier (RGI) tool available on the Comprehensive Antibiotic Resistance Database website (CARD 3.2.4; <https://card.mcmaster.ca>) (McArthur et al., 2013). RGI utilizes a combination of software including Prodigal for open reading frame prediction (Hyatt et al., 2010) and sequence alignment tools such as BLAST (Altschul et al., 1990) or DIAMOND (Buchfink et al., 2015) to identify ARGs present in uploaded genome sequences by comparing them to the profile of ARGs within the CARD database. The CARD profile of ARGs include curated antibiotic resistance mutations allowing RGI to detect ARG-associated mutations directly from genomic data.

In order to classify genes into different "antibiotic drug classes", we set the BLAST sequence identity threshold to 95%. Subsequently, we counted the number of genes representing various resistance mechanisms for each strain of *A. baumannii* genomes. Using an in-house Perl script, we constructed an abundance matrix of ARGs for all the 223 *A. baumannii* genomes, where the columns denoted the resistance mechanisms and the rows represented the *A. baumannii* genomes.

2.5. Prediction of virulence factors (VFs)

To identify potential VFs in the curated *A. baumannii* genomes, an *in-silico* analysis was conducted searching against the Virulence Factor Database (VFDB). VFDB is a manually curated database of known and predicted bacterial VFs derived from various pathogen genomes and Transcriptomic data (Chen et al., 2005).

The analysis began by extracting 135 protein sequences of experimentally verified VFs that originated specifically from *A. baumannii* from the VFDB core dataset. These 135 protein sequences were used to construct a local BLASTP database to serve as references for the analysis. Each of the 223 curated *A. baumannii* proteomes was then searched against this database using BLASTP with stringent parameters of 95%

identity and 95% query coverage. This step aimed to identify homologs of known *A. baumannii* VFs present in each genome.

Finally, the BLASTP results were parsed and loaded into an in-house script to generate a 135×223 abundance matrix. This output matrix describes the presence or absence (expressed as count values) of each virulence factor across all 223 genomes. Together, these steps allowed prediction of potential virulence profiles for each *A. baumannii* strain based on alignment to validated reference VFs.

2.6. Phylogenetic analysis

The phylogenetic relationships between the 223 *A. baumannii* strains were inferred through construction of a core genome phylogenetic tree. First, the core genome alignment file containing single-nucleotide polymorphism (SNP) sites across all core protein coding sequences, as identified previously through Roary analysis, was obtained.

This alignment file was loaded into the molecular evolutionary genetics analysis (MEGA) software version 11 (Tamura et al., 2021). The nucleotide sequences from the core genome alignment were used to reconstruct the phylogenetic tree in MEGA11. Specifically, the neighbor-joining (NJ) algorithm was applied to build the initial phylogenetic network based on nucleotide distance calculations between sequences (Wu et al., 2006).

To determine the statistical robustness and reliability of branching orders within the tree, bootstrap resampling was performed, utilizing 500 replicate runs. This assigned confidence values to the tree nodes based on the percentage of bootstrap resampling that supported each branching cluster. The resulting phylogenetic tree with bootstrap support values visualized the genetic relationships between the strains based on their core genome SNP profiles. It provided insights into the mutational divergence and population structure of the *A. baumannii* collection.

2.7. Statistical analysis

To identify the inter- and intra-cluster (representing major STs detected in our study) variations in the abundances of ARGs and VFs, we conducted Welch's *t*-test. The statistical significance was assessed at $p < 0.05$.

3. Results

3.1. MLST analysis

By applying the Oxford scheme for MLST analysis, 90 distinct sequence types (STs) were identified within 212 out of the total 223 genomes (Maiden et al., 2013). However, the STs for the remaining 11 genomes could not be determined. Out of the 90 different STs found, 64 were linked to single *A. baumannii* genomes, while the remaining 26 characterised multiple *A. baumannii* strains. Remarkably, ST208, ST191, and ST195 stood out as the most prevalent STs (Supplementary Table 2). Among the 212 genomes, 35 (17%) were associated with ST208, 16 (8%) with ST191, and 15 (7%) with ST195. Additionally, ST368 and ST865 were represented by 7 and 9 strains, respectively. The remaining STs were found in only five or lesser number of *A. baumannii* strains.

3.2. Pangenome profiling of *A. baumannii* genomes

The BV-BRC analysis of 223 *A. baumannii* strains showcased a range of protein-coding sequences, spanning from 3449 to 5524 on average 3931.92 ± 214.05 . (Supplementary Table 1). The average GC-content of all 223 genomes was found to be $39.02\% \pm 0.44$. 28,516 clusters encompassing 853,965 protein-coding genes were identified, forming putative orthologous sets across various genomes, commonly referred to as the pan-genome. The core genome consists of 1305 genes, which can be further divided into two sets: Hard Core (617 genes) and Soft Core

(688 genes), representing genes found in 100% and 99% of the genomes, respectively. Furthermore, the identification of over 24,000 cloud genes (singletons or genes exclusive to a single genome) underscores considerable diversity in gene content among *A. baumannii* genomes, aligning with prior findings (Sahl et al., 2015, Urhan and Abeel, 2021) (Supplementary Fig. 1).

Heap's Law was employed to calculate the openness and closeness status of the *A. baumannii* pangenome, following the methodology outlined by Tettelin et al. (Tettelin et al., 2008). According to their findings, the pan-genome's nature, whether finite or infinite, can be predicted using Heaps' law and a power law. Heaps' law is expressed as $n = \kappa N^\gamma$, where n is the pan-genome size, N is the number of genomes utilized, and κ and γ are the fitting parameters. The exponent γ determines the openness of the pan-genome; $\gamma < 0$ indicates a closed pan-genome, where size stabilizes with additional genomes, while $\gamma > 0$ suggests an open pan-genome with size expanding as more genomes are incorporated. The open property of the pan-genome was also assessed using a power law model with the formula $\Delta n = \kappa N^{-\alpha}$, where Δn is the number of newly added genes, N is the number of genomes, and κ and α are fitting parameters. For $\alpha > 1$, the pan-genome is considered closed, and for $\alpha < 1$, it is deemed open. Notably, Heaps' law and the power law exhibit mathematical similarity. The fitting of 2000 random permutations of all *A. baumannii* genomes against Heap's Law resulted in a λ value of 0.391 ± 0.001 , indicative of an open pangenome (Fig. 1A). The observation that the pan-genome size increased proportionally with the expansion of genome numbers supports the conclusion that the *A. baumannii* pan-genome is capable of acquiring new genes (Imperi et al., 2011).

3.3. Phylogenetic analysis and selection of orthologous gene clusters

Roary was used to generate a core gene alignment, which was utilized for constructing the core phylogeny (Fig. 2) comprising 223 *A. baumannii* genomes. The core phylogeny revealed 25 distinct clusters, each labeled with the identified STs of the genomes. Every node in the core phylogeny exhibited a bootstrap value equal to or exceeding 70%, indicating strong support for the inferred relationships among the taxa represented by those nodes. Notably, the five predominant sequence types (ST208, ST195, ST368, ST191, and ST865) identified through MLST analysis were distributed across five distinct clusters within the core phylogeny, with each cluster encompassing more than five genomes. Designated as Cluster 1 through Cluster 5, these strains collectively represented 47% of the analysed *A. baumannii* strains. The remaining 53% strains were typed as XX number of diverse STs. Our subsequent analyses focused exclusively on Cluster 1 through Cluster 5, which are detailed in Table 1, including the major sequence types (STs) associated with them and their corresponding geographic locations. Apart from ST208, the remaining four major STs are confined within their individual clusters in the core phylogeny. Among the five major STs, except ST208, the other four STs are from Asian countries like India, China and South Korea. The probable STs for the 11 *A. baumannii* genomes with unknown STs can be inferred by examining the positions of neighboring genomes within the same cluster in the core phylogeny. (Supplementary Table 3).

It has been found that with each successive addition of strains, the pan-genome expanded by an average of about 110 genes, ultimately culminating in the identification of 28,516 genes. Steeper increases were observed between clusters, while the plateau was evident within clusters (Fig. 1A). We conducted an additional analysis on the core (genes shared by all genomes), shell (genes shared by multiple genomes but not all), and cloud gene (genes present in only one genome) counts within the five primary clusters. Except for cluster 1, the major STs exhibited a greater count of core genes in comparison to shell and cloud genes (Fig. 3A). Clusters 5 and 3 displayed the most notable distinctions between the core and shell gene counts, a finding substantiated by the chi-squared test. This observation implies that although *A. baumannii* strains

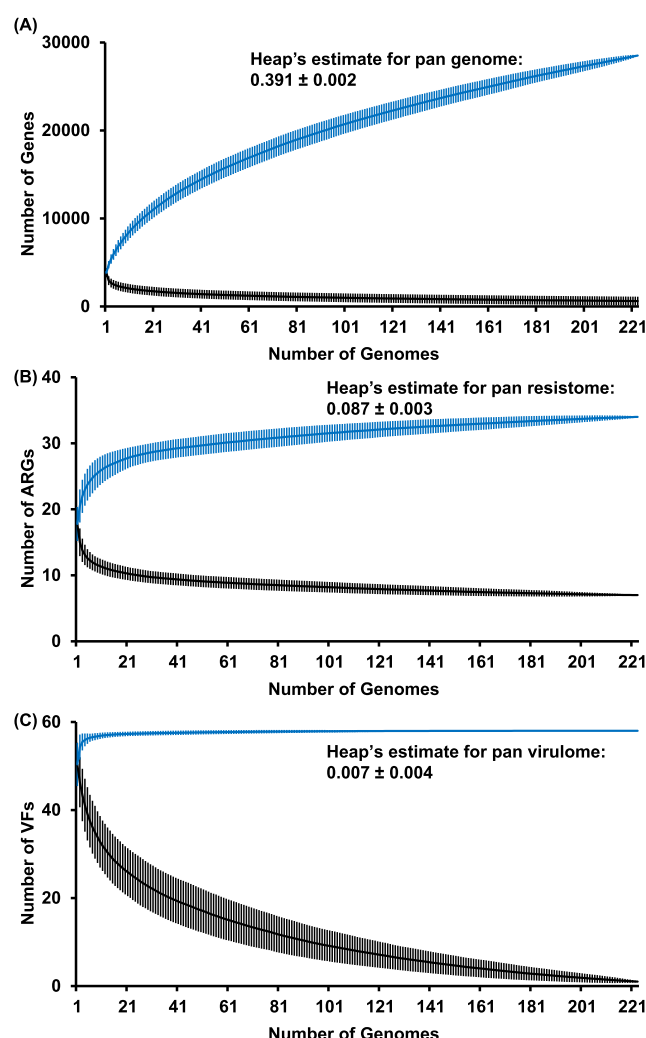


Fig. 1. (A) Graph representing the pan-genome (blue) and core-genome (black) of the 223 analysed *A. baumannii* genomes. (B) Graph representing the pan-resistome (blue) and core-resistome (black) of the 223 analysed *A. baumannii* genomes. (C) Graph representing the pan-virulome (blue) and core-virulome (black) of the 223 analysed *A. baumannii* genomes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

exhibit an expansive pan-genome, the core genes among the five major STs tend to remain conserved, potentially contributing to distinct pan-genome stabilizations.

3.4. Antibiotic resistance genes (ARGs) profiling

The ARGs in the curated 223 *A. baumannii* genomes were determined using the RGI tool. Based on the CARD database, RGI provided predictions regarding the antibiotic drug classes against which the genes are resistant to. This analysis revealed that *A. baumannii* isolates presented the most resistance towards the single drug-class “aminoglycoside antibiotic”. We further analysed the ARG profiles of genomes from the five selected clusters from phylogenetic analysis (see section 3.3). A total of 34 different drug classes, comprising 18 multi-drug classes and 16 single-drug classes, were predicted for the *A. baumannii* genomes in this study (Supplementary Table 4A). It should be noted here that the nomenclature of the drug classes that the genes are resistant against are taken exactly as expressed in the CARD database and have not been modified.

3.4.1. Inter-cluster variations in relation to ARGs

We aimed to identify significant differences in ARGs among the five selected clusters. Out of the 34 drug classes predicted by RGI, only 16 drug classes were identified, which exhibited inter-cluster variations (The cells under the column “ARG types” have been marked in yellow in Supplementary Table 4B). Notably, the ‘aminoglycoside antibiotic’ and ‘sulphonamide antibiotic’ drug classes (The text of the cells under the column “ARG types” have been marked in red in Supplementary Table 4B) showed the highest variation representing 6 or more significant inter-cluster variations (>50% of the 10 possible inter-cluster combinations). This indicates, for the five selected clusters, there are significant differences in the number of ARGs against the ‘aminoglycoside antibiotic’ and ‘sulphonamide antibiotic’ drug classes. Additionally, the occurrence of ARGs against the multi-drug class ‘monobactam; cephalosporin; penam; penem’ also displayed substantial variation among the major STs representing 5 significant inter-cluster variations (the text of the cells under the column “ARG types” have been marked in green in Supplementary Table 4B) (Fig. 5).

Given that the five primary clusters (section 3.3) correspond to the five major STs for the 223 *A. baumannii* genomes, the variations observed between these clusters concerning ARGs undoubtedly signify the diversity of ARGs among the major STs. ST208 and ST865 showed significantly higher numbers ($P < 0.05$) of ARGs compared to the other four major STs (Fig. 4A). ST208 is present in two clusters viz., Cluster 1 and Cluster 3 (Supplementary Table 1). Interestingly, these two separate groups of ST208 displayed similar resistant drug-class distributions, differing only in the presence of ARGs against a broad range of drug classes like macrolide; fluoroquinolone; lincosamide; carbapenem; cephalosporin; tetracycline; rifamycin; diaminopyrimidine; phenicol and penem (the text of the cells under the column “ARG types” have been marked in pink in Supplementary Table 4B & Fig. 5).

3.4.2. Intra-cluster variations in relation to ARGs

Comparing the distribution of the ARGs for the major STs present in a cluster with that of the other STs present in the same cluster, we observed that the differences were not as significant as the inter-cluster variations. This suggests that the gain and loss of ARGs in *A. baumannii* genomes occurred as a result of their core genome diversification. For Cluster 3, containing two major STs (ST208 and ST368), significant variation was noticed for the ARGs against aminoglycoside antibiotic between ST208 and ST368 (Supplementary Table 4C).

3.4.3. Pan-resistome profiling of *A. baumannii* genomes

Openness and closeness status of the *A. baumannii* pan-resistome (the entire set of ARGs) was calculated using Heap’s Law (similar approach was followed as mentioned in section 2.3). 2000 random permutations of all the *A. baumannii* resistomes were run for fitting it against Heap’s Law which generated a λ value of 0.087 ± 0.003 indicating an open pan-resistome and the core resistome was found to be closed (Fig. 1B). We further computed the number of core-ARGs, shell-ARGs, and cloud-ARGs for each of the STs, considering the five major clusters. Across all the clusters, each of the five major STs consistently exhibited a higher number of core-ARGs compared to the shell and cloud ARGs. The differences between the number of core and shell ARGs for each of the five major STs were most pronounced in cluster 5 and cluster 3, as evident from the chi-squared test results (Fig. 3B). Notably, the evolution of *A. baumannii*’s resistome follows a pattern similar to its pan-genome evolution.

3.5. Virulence factor profiling

We found a total of 58 types of putative VFs in *A. baumannii* genomes under study, which could be classified into five major types: exotoxin, immune modulation, biofilm, ‘regulation’ of signal transduction system BfmR/BfmS, and nutritional/metabolic factors (Liu et al., 2022) (Supplementary Table 5 A). Each of the 223 isolates possessed a range of 25

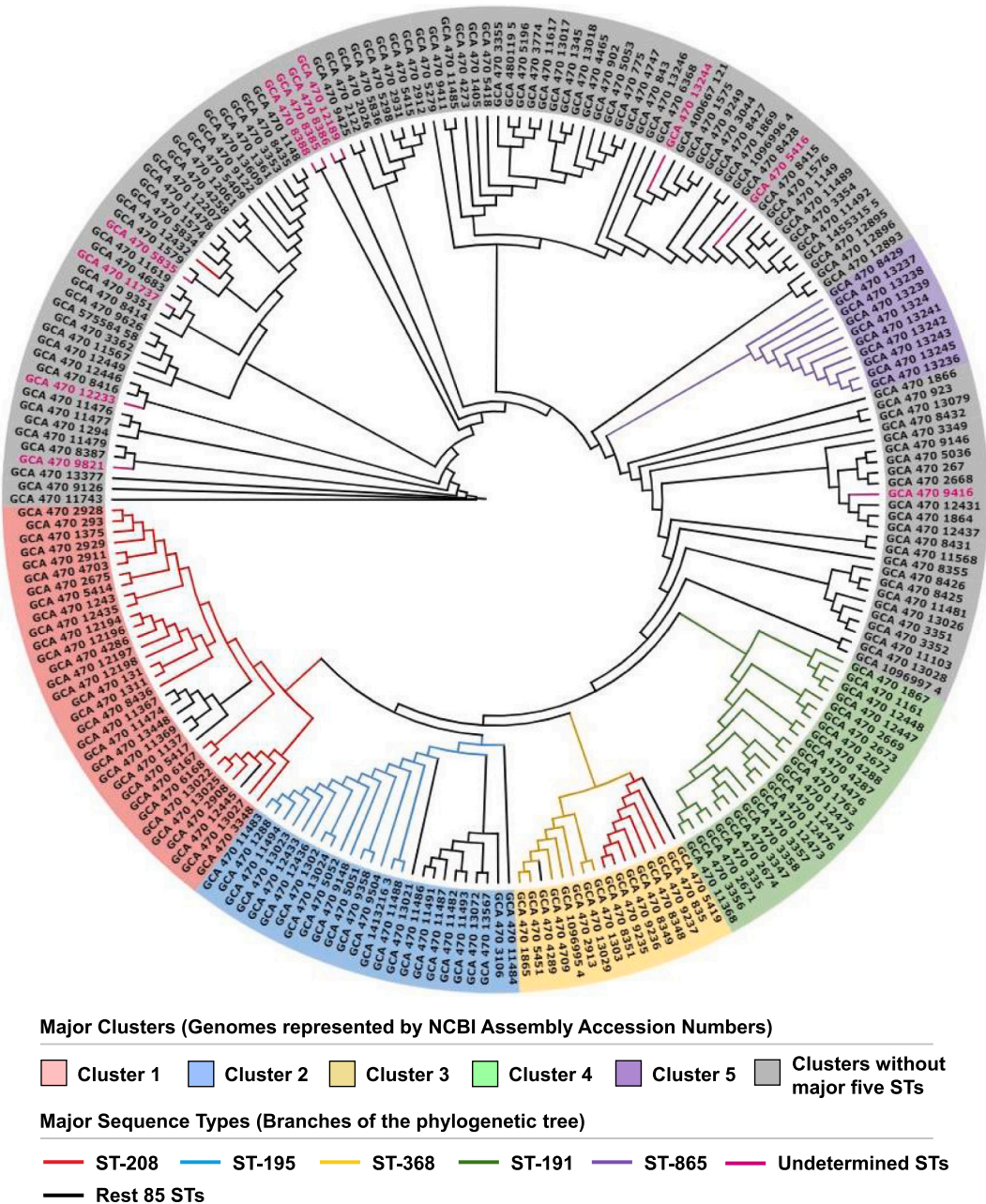


Fig. 2. Core phylogeny of the 223 *A. baumannii* genomes. Five clusters representing five major STs have been coloured.

to 58 VFs per genome, with an average value 50.32 ± 4.94).

3.5.1. Variations in VFs among and within clusters

We examined the differences in the quantities of VFs among the five principal STs of *A. baumannii*. Exotoxin and nutritional/metabolic factors emerged as key contributors in distinguishing the five major STs (Supplementary Table 5B). Within Cluster 3, the distribution of VFs between ST208 and ST368 was significantly impacted by the presence of genes related to exotoxin, immune modulation, and biofilm (Supplementary Table 5C). Moreover, the pivotal roles of exotoxin and biofilm became evident in distinguishing the distribution patterns of VFs between the two subsets of ST208 found in Cluster 1 and Cluster 3. (Supplementary Table 5B). The gain and loss of VFs seem to influence the diversification of major STs of *A. baumannii* strains (Fig. 4B). Specifically, ST208 isolated from Canada in Cluster 3, along with other members of Cluster 1 and Cluster 3, showed a lower abundance of VFs, particularly biofilm-related VFs. These findings highlight the role of VFs

in shaping the genomic diversity of *A. baumannii* strains.

3.5.2. Pan-virulome profiling of *A. baumannii* genomes

Openness and closeness status of the *A. baumannii* pan-virulome (the entire set of VFs) was also calculated using Heap's Law (similar approach was followed as mentioned in section 2.3). 2000 random permutations of all the *A. baumannii* virulomes were run for fitting it against Heap's Law which generated a λ value of 0.007 ± 0.004 indicating a not very much open pan-virulome (Fig. 1C). According to Antunes et al., the core genome of *A. baumannii* encompasses the majority of well-known VFs. (Antunes et al., 2014). This observation suggests that essential elements contributing to *A. baumannii*'s virulence are inherent and relatively stable across strains, reinforcing the concept of a not much open pan-virulome of *A. baumannii*. Though the core-virulome was closed in nature, but it was found to contain a much variable set of genes among different strains of *A. baumannii*.

We conducted calculations to determine the number of core-VFs,

Table 1
Details of five analysed clusters in core phylogeny and distribution of major STs.

Cluster no.	No. of genomes present in the cluster	Geographical location of the genomes	Major STs of the cluster	No. of genomes with major STs	Geographical location of the genomes with major STs
1	32	North America (USA, Canada, and Mexico), Asia (China and South Korea), Australia	ST208	25	China, USA, Mexico, Australia
2	24	Asia (India, China, Malaysia, Thailand, Turkey, Lebanon)	ST195	15	India, China
3	16	Asia (China), North America (Canada)	ST208, ST368	7, 7	Canada (ST208), China (ST368)
4	23	Asia (South Korea, China), USA	ST191	16	Asia (South Korea, China)
5	10	Asia (India)	ST865	9	Asia (India)

shell VFs (genes shared by more than one genome but not all), and cloud VFs for each of the STs, considering the five major clusters. For all the five clusters and the five major STs, we consistently observed a higher

number of core-VFs compared to the shell and cloud VFs (Fig. 3C). The differences in the number of core-VFs and shell VFs (shell and cloud VFs) among the five major STs were most pronounced for clusters 5, 3, and 2, as indicated by the results of the chi-squared test. But, comparison of Fig. 3B and C indicates that for most of the STs, the number of core-VFs are much higher than core-ARGs.

4. Discussions

The present study aimed towards a thorough comparative genomic analysis of 223 curated genomes of *A. baumannii*, deciphering their genetic diversity with a specific focus on antibiotic resistance and VFs. Applying the Oxford MLST scheme to 223 *A. baumannii* genomes, we found 90 distinct STs, with 64 linked to single genomes and 26 to multiple genomes. Notably, ST208, ST191, and ST195 were the most prevalent (Supplementary Table 2). Pan-genome analysis of the selected *A. baumannii* genomes, revealed 1305 genes in the core genome, and the detection of over 24,000 cloud genes emphasizes significant gene content diversity in *A. baumannii* genomes, consistent with previous research. Touchon et al. have also found extensive pan-genome of *A. baumannii* with highly varied gene repertoires, implying frequent horizontal gene transfer. They emphasized that *A. baumannii* originated from an ancient population bottleneck, followed by population expansion under strong purifying selection (Touchon et al., 2014). The openness and closeness of the *A. baumannii* pan-genome were assessed using Heap's Law. Fitting all *A. baumannii* genomes against Heap's Law indicated an open pangenome. As the number of genomes increased, the pan-genome size also increased, suggesting continuous acquisition of

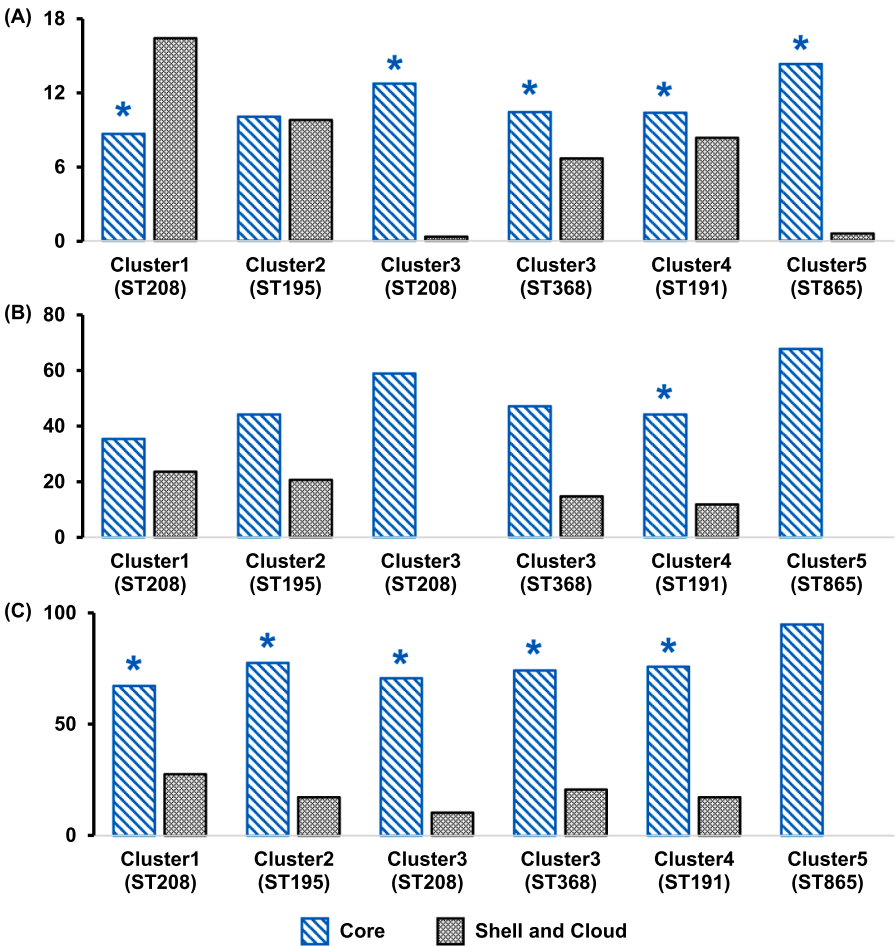


Fig. 3. Comparison of core gene count with shell and cloud gene count for five major STs in (A) Genome (B) Resistome (C) Virulome of the 223 *A. baumannii* genomes. * denotes statistical significance ($P<0.05$) of the differences between two groups.

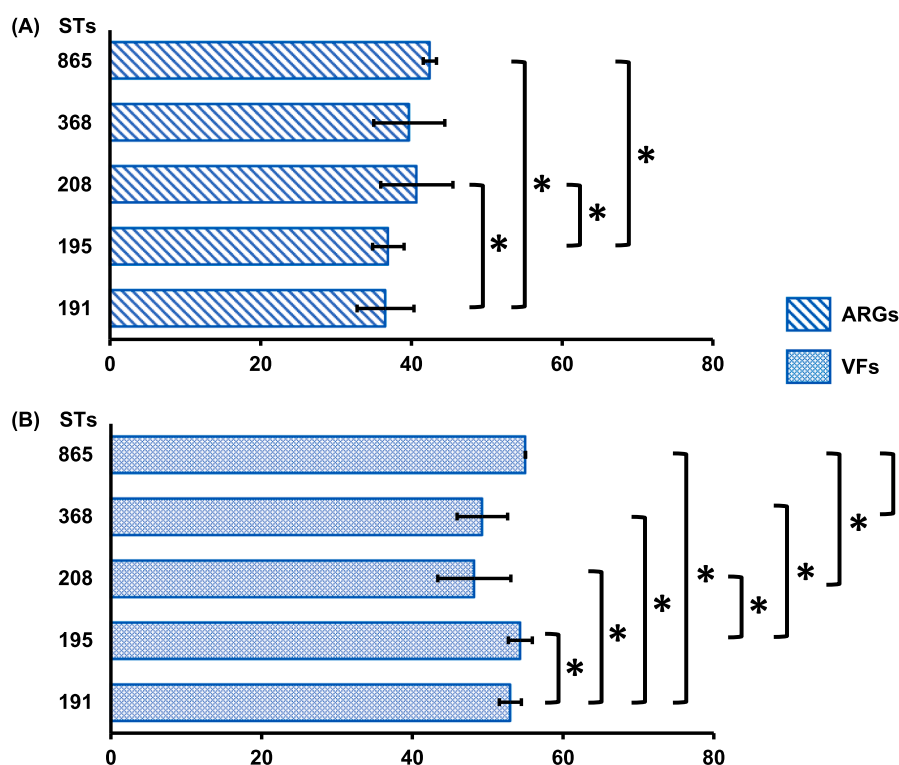


Fig. 4. Comparison of the ARGs (A) and VFs (B) count in five major STs of the 223 *A. baumannii* genomes. * denotes statistical significance ($P < 0.05$) of the differences between two groups.

new genes in the *A. baumannii* pan-genome. Conversely, a diminishing trend was observed in the core genome size across iterations (Fig. 1A).

The core phylogeny revealed 25 distinct clusters, and the five major STs (ST208, ST195, ST368, ST191, and ST865) from MLST analysis were distributed across five clusters, collectively representing 47% of *A. baumannii* genomes (Fig. 2). Except for ST208, the other four major STs were confined to individual clusters in the core phylogeny, originating from Asian countries like India, China, and South Korea.

Profiling of ARGs of the 223 *A. baumannii* isolates showed heightened resistance to aminoglycoside antibiotics. Analysing genomes from five clusters, we predicted 34 drug classes, only 16 of them exhibited inter-cluster variations. Notably, ‘aminoglycoside antibiotic’ and ‘sulphonamide antibiotic’ classes showed the highest variation, indicating their significant differences among the major STs. Furthermore, the presence of ARGs against the multi-drug class ‘monobactam; cephalosporin; penam; penem’ also showed significant variation among the major STs (Supplementary Table 4B). ST208 is identified in both Cluster 1 and Cluster 3. Notably, these distinct ST208 groups exhibited comparable distributions of resistant drug classes, differing only in the presence of ARGs against multiple drug-classes such as macrolide, fluoroquinolone, lincosamide, carbapenem, cephalosporin, tetracycline, rifamycin, diaminopyrimidine, phenicol, and penem. Comparing ARG distribution within major STs in a cluster to other STs in the same cluster (intra-cluster) showed less significant differences than inter-cluster variations (Supplementary Table 4C). This indicates that ARG gain and loss in *A. baumannii* genomes result from core genome diversification. The global dissemination of carbapenem resistance in *A. baumannii* strains is well-known, and our findings suggest that the gain and loss of ARGs against carbapenem are correlated with the genomic diversities of *A. baumannii* isolates belonging to ST208 (Bartal et al., 2022). Overall, our results suggest that major five STs (Table 1) of *A. baumannii* exhibit distinct variations in their ARG distribution, with a prominent role played by the ARGs against aminoglycosides and sulphonamide in segregating the major STs (Fig. 5). ARGs against aminoglycoside,

macrolide, fluoroquinolone, lincosamide, carbapenem, cephalosporin, tetracycline, rifamycin, diaminopyrimidine, phenicol, penem and sulphonamide were found to be most prevalent for the five major STs of *A. baumannii* genomes (Table 1).

In the present study, we have found both the geographical isolation sites and STs orientation influenced the distribution pattern of resistant drug classes. Resistant genes against ‘phenicol antibiotic’ were found to be missing in the isolates from USA (ST208), which are present in Cluster 1, whereas, resistant genes against ‘monobactam, cephalosporin, penam, penem, phenicol and sulphonamide’ were found to be missing for the two isolates from the same Cluster whose isolation area is unknown. The rest of the ST208 members of Cluster 1 (isolated from China), were found to contain all the five drug class categories mentioned in the previous paragraph. Hence, the distribution pattern of ARGs among *A. baumannii* isolates in Cluster 1 was observed to be more affected by geographical variations than their STs (Fig. 7 & Supplementary Table 6). Members within Cluster 2, specifically those of ST195 (all obtained from Asia), possess a lower abundance of ARGs in comparison. They entirely lack antibiotic ARGs targeting phenicols. Furthermore, some of them lack ARGs conferring resistance to monobactam, cephalosporin, penam, and penem, while others are deficient in sulfonamide-resistant genes (Fig. 7 & Supplementary Table 6). Within Cluster 3, there exist two STs: ST208 and ST368. Individuals belonging to ST368 exhibit a lower count of ARGs, lacking at least one among the five ARGs (Fig. 5). Conversely, members of ST208, isolated from Canada within Cluster 3, demonstrate a higher abundance of ARGs. Hence, when examining two distinct sets of ST208—one from Cluster 1 and the other from Cluster 3—we identified diverse patterns of ARG distribution based on their geographical locations. Certain individuals within Cluster 4 (ST191), primarily isolated from South Korea, lack ARGs associated with antibiotic drug classes such as monobactam, cephalosporin, penam, and penem (Fig. 7 & Supplementary Table 6). Additionally, a few members within this cluster are deficient in ARGs against phenicols. Cluster 5 (ST865) isolates from India were also found to lack ARGs targeting drug

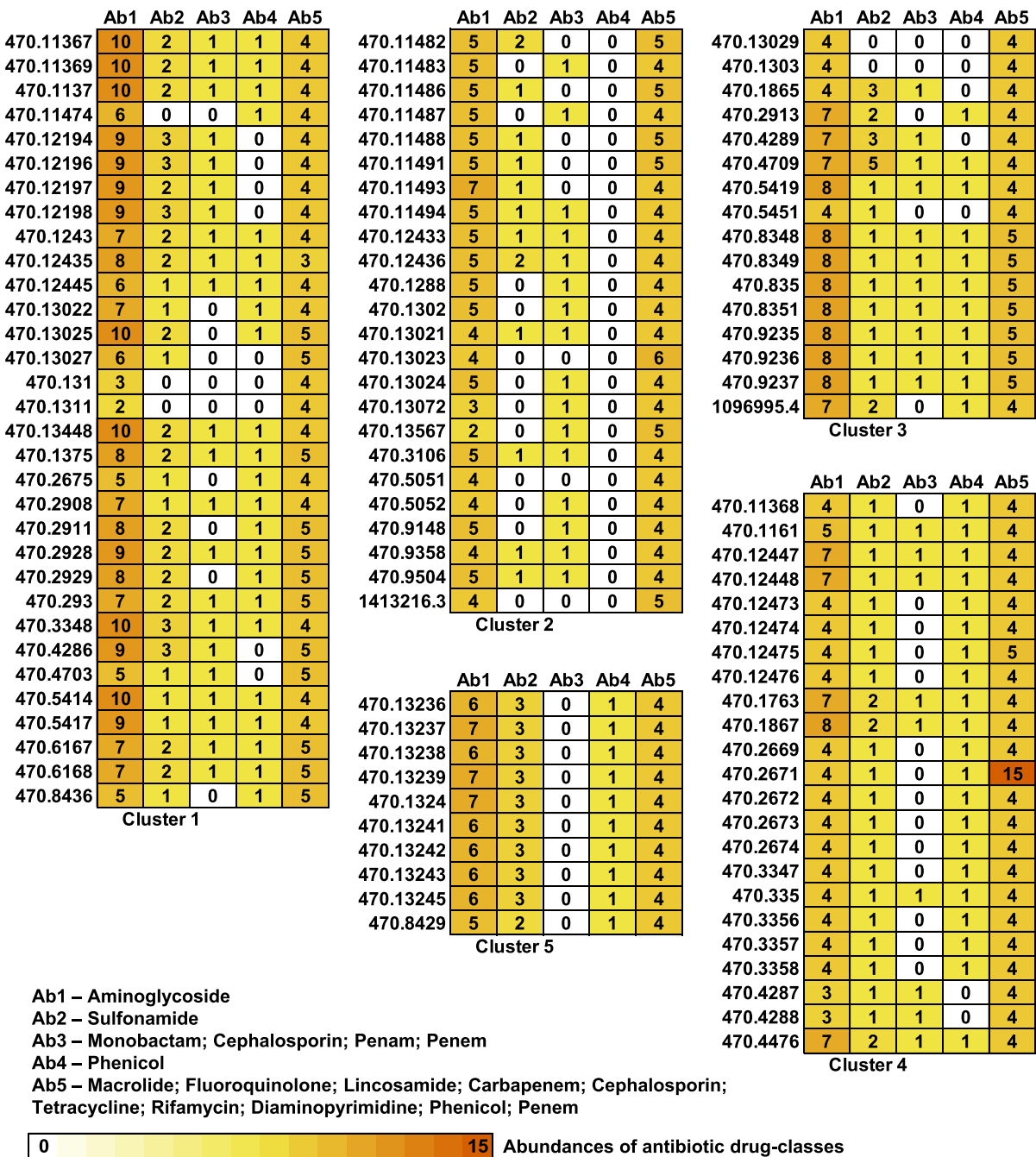


Fig. 5. Abundances of key antibiotic drug-classes having prominent roles in segregating the major STs of the 223 *A. baumannii* genomes.

classes such as monobactam, cephalosporin, penam, and penem (Fig. 7 & Supplementary Table 6). Therefore, it can be inferred that ST195 and ST365, both obtained from India, exhibit variations in the distribution pattern of their ARGs based on their core phylogeny distribution. The pan-resistome of *A. baumannii* isolates was determined to be open, implying ongoing diversity and the potential for new resistance genes with the addition of more samples. On the other hand, the core resistome, which likely represents a set of conserved resistance genes across the population, was found to be closed. This suggests that the core ARGs in *A. baumannii* are relatively stable and less influenced by the addition of more resistomes.

Understanding the interactions of VFs is essential for developing effective strategies against *A. baumannii* infections (Moubareck and Halat, 2020). Exotoxin and nutritional/metabolic factors played crucial roles in differentiating the five major STs (Supplementary Table 5B). In

Cluster 3, the distribution of VFs between ST208 and ST368 was notably influenced by genes related to exotoxin, immune modulation, and bio-film (Supplementary Table 5C, Fig. 6). Additionally, the distinctive roles of exotoxin and biofilm were evident in discerning the VF distribution patterns between the two subsets of ST208 within Cluster 1 and Cluster 3. *A. baumannii* strains cause extensive infections and rapidly acquire antibiotic resistance, posing a significant healthcare threat (Fig. 6).

A. baumannii adapts to hostile host environments by altering its metabolism and nutritional requirements (Mortensen and Skaar, 2013). The nutrient requirements may vary among strains, and the host's defence mechanisms can impede the process of nutrient acquisition. Moreover, exotoxins, such as phospholipases C and D, are well-known VFs in *A. baumannii* (Schmiel and Miller, 1999). These exotoxins are substrate-specific towards phosphatidylcholine, aiding in iron acquisition through their haemolytic activity against human erythrocytes

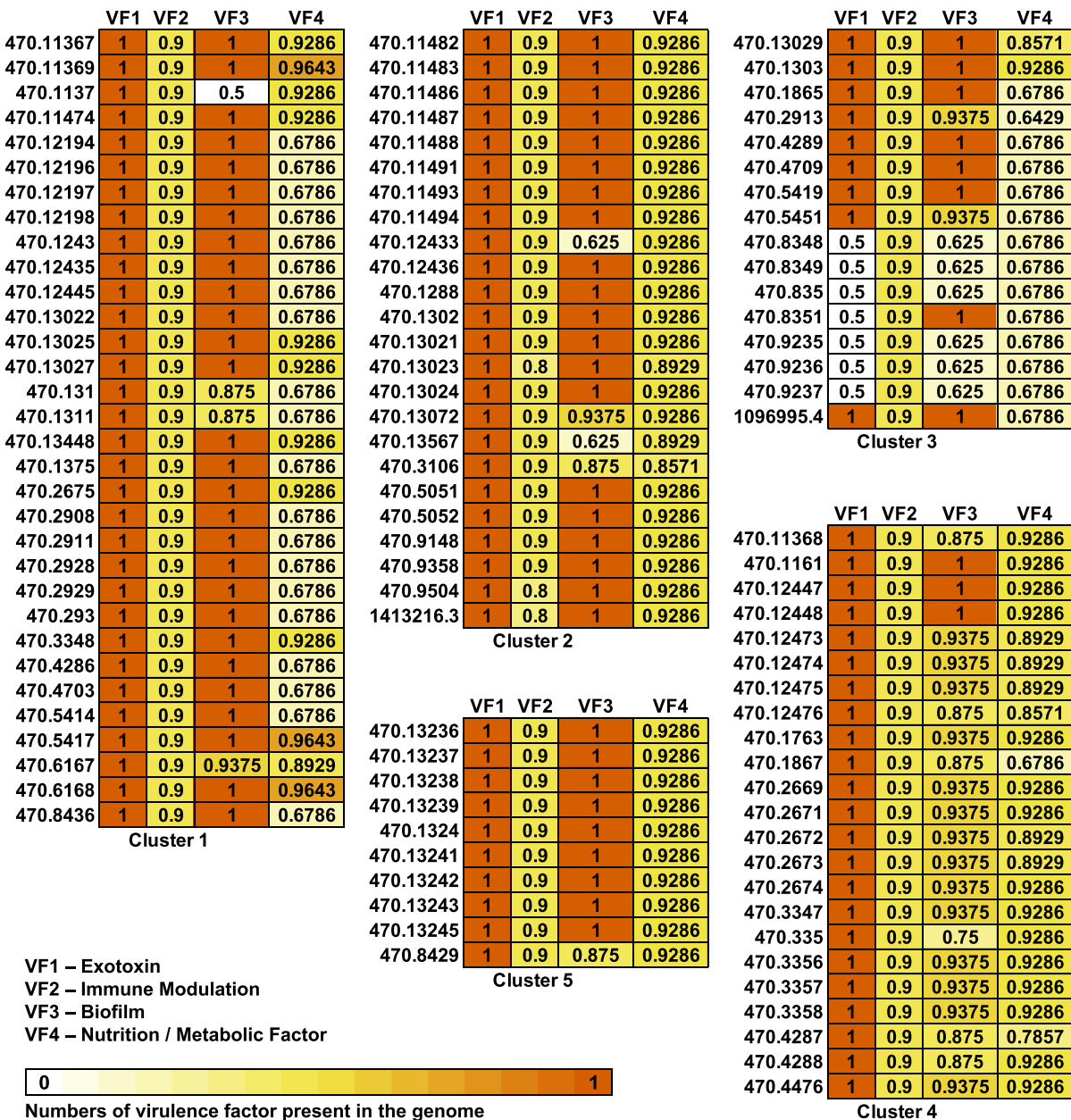


Fig. 6. Abundances of key virulence factors having prominent roles in segregating the major STs of the 223 *A. baumannii* genomes.

(Chadha et al., 2022; Fisher et al., 2018). *A. baumannii* strains are also adept at biofilm production, allowing them to adhere to surfaces like clinical instruments, endotracheal tubes, and catheters. Biofilms are structured bacterial communities embedded within a self-produced matrix. They serve as potent VFs for *A. baumannii*, contributing to increased antibiotic and antimicrobial stressor resistance (Gedefie et al., 2021; Dolma et al., 2022). The utilization of these VFs enhances *A. baumannii*'s ability to survive and cause infections in diverse environments, contributing to its success as a pathogen and presenting challenges for effective treatment and control measures. The variation in VFs appears to contribute to the diversification of major STs in *A. baumannii* strains. Notably, ST208 from Canada in Cluster 3, along with members of Cluster 1 and Cluster 3, exhibited a reduced abundance of VFs, particularly those related to biofilm (Fig. 7 & Supplementary Table 6). These results underscore the significance of VFs in influencing the genomic diversity of *A. baumannii* strains. This implies that crucial elements contributing to *A. baumannii*'s virulence are inherent and relatively stable across strains, supporting the idea of a relatively limited

pan-virulome. While the core-virulome is closed, it contains a variable set of genes among different *A. baumannii* strains.

5. Conclusion

In conclusion, our study sheds light on the concerning prevalence of MDR *A. baumannii* isolates, particularly displaying resistance to aminoglycosides, glycolcycline, and tetracycline. ST208 was found to dominate among the various STs, with clear geographical orientations observed. The open pan-genome of *A. baumannii* contributes to its genomic versatility and adaptability, as reflected in the diversification of the core genome with the gain and loss of antibiotic resistance and virulence genes. We observed a versatile resistome in *A. baumannii*, indicating a widespread occurrence of strains resistant to multiple classes of antibiotic drugs (Fig. 7). This poses a significant threat to treatment options, urging the need for effective strategies to combat this pathogen and preserve antimicrobial efficacy. Though the presence of some unclassified STs is a limitation, the study has successfully analysed

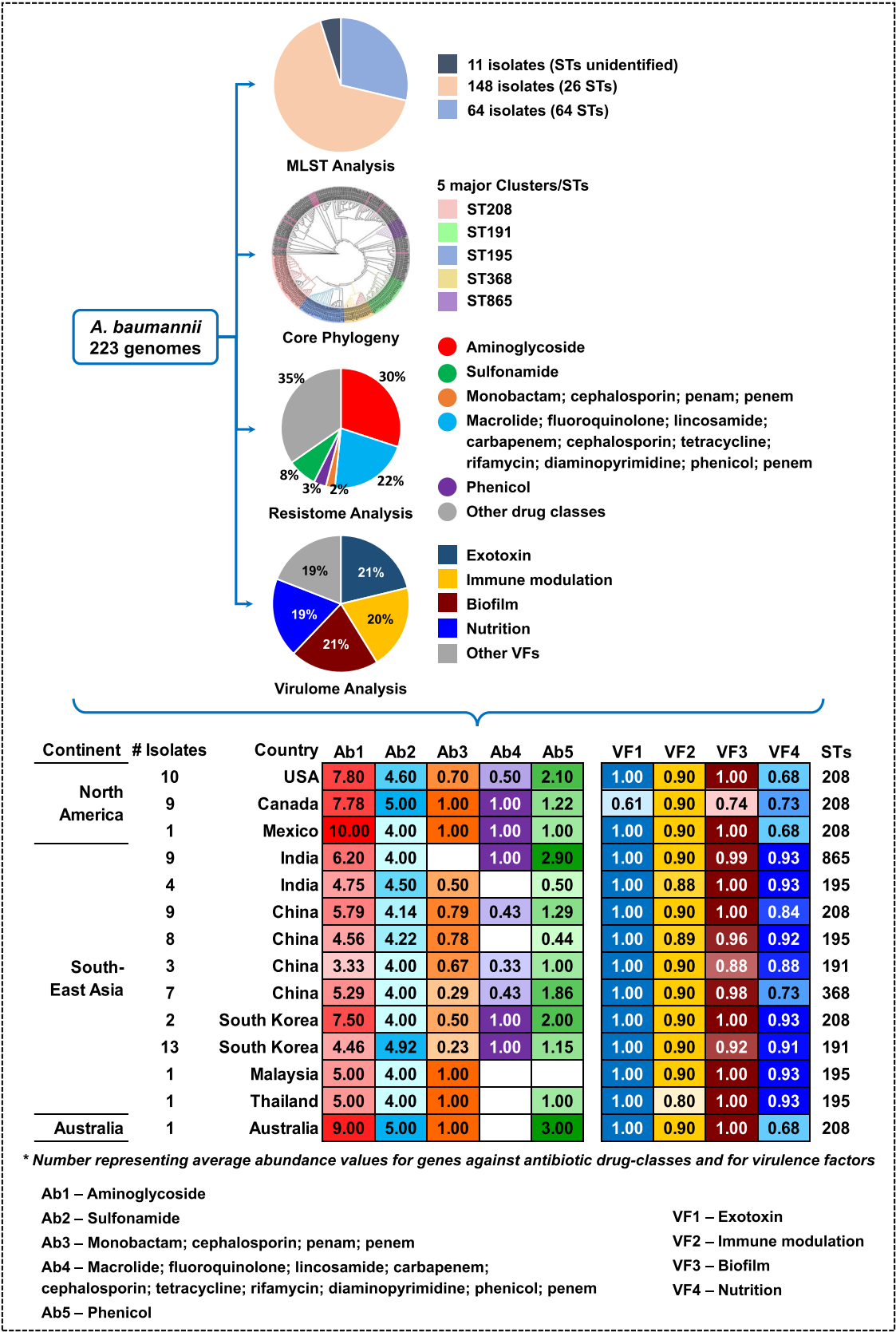


Fig. 7. Schematic representation illustrating the correlation among ARGs, VFs, and the five major phylogenetically determined STs within the core genomes across 223 *A. baumannii* genomes.

the pan-genome, resistome, and virulome of major STs of considerable numbers of *A. baumannii* strains. ST865 showed the most stable pan-genome, resistome, and virulome, offering insights into potential targets for therapeutic interventions.

By enhancing our understanding of the genetic basis of MDR and virulence in *A. baumannii* worldwide, our findings may contribute to the development of diagnostic tools, therapies, and preventive measures. Addressing the challenge of *A. baumannii* as a pathogen requires concerted efforts and alignment with current scientific approaches. We hope that our study will aid in combating *A. baumannii* infections and limiting the spread of MDR strains, ultimately contributing to improved healthcare outcomes.

Author contributions

SP and RB conceptualized the project, performed the analysis, generated and organized the results. SMR, AL, and PV carried out data retrieval, data pre-processing and initial analysis. AKB, SB and KB performed required programming during analysis and data generation. RB conceived and coordinated the project, performed data analysis and interpretation. SP and RB wrote the manuscript with inputs from other authors. All the authors read and approved the final manuscript.

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Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

All data that were used in this research work, are publicly available. Details of these data are mentioned in Supplementary Table 1.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2024.105579>.

References

- Aguepong, N., Fordjour, F., Owusu-Ofori, A., 2023. Multidrug-resistant *Acinetobacter baumannii* in healthcare settings in Africa. *Front. Trop. Diseases* 4, 1110125.
- Aliramezani, A., Soleimani, M., Fard, R.M., Nojoomi, F., 2019. Virulence determinants and biofilm formation of *Acinetobacter baumannii* isolated from hospitalized patients. *Germes* 9 (3), 148.
- Altschul, S.F., Gish, W., Miller, W., Myers, E.W., Lipman, D.J., 1990. Basic local alignment search tool. *J. Mol. Biol.* 215 (3), 403–410.
- Antunes, L.C., Visca, P., Townner, K.J., 2014. *Acinetobacter baumannii*: evolution of a global pathogen. *Pathog. Diseases* 71 (3), 292–301.
- Bartal, C., Rolston, K.V., Neshet, L., 2022. Carbapenem-resistant *Acinetobacter baumannii*: colonization, infection and current treatment options. *Infect. Dis. Ther.* 11 (2), 683–694.
- Beceiro, A., Tomás, M., Bou, G., 2013. Antimicrobial resistance and virulence: a successful or deleterious association in the bacterial world? *Clin. Microbiol. Rev.* 26 (2), 185–230.
- Boucher, H.W., Talbot, G.H., Bradley, J.S., Edwards, J.E., Gilbert, D., Rice, L.B., Scheld, M., Spellberg, B., Bartlett, J., 2009. Bad bugs, no drugs: no ESAP! An update from the Infectious Diseases Society of America. *Clin. Infect. Dis.* 48 (1), 1–2.
- Buchfink, B., Xie, C., Huson, D.H., 2015. Fast and sensitive protein alignment using DIAMOND. *Nat. Methods* 12 (1), 59–60.
- Chadha, J., Harjai, K., Chhibber, S., 2022. Revisiting the virulence hallmarks of *Pseudomonas aeruginosa*: a chronicle through the perspective of quorum sensing. *Environ. Microbiol.* 24 (6), 2630–2656.
- Chen, L., Yang, J., Yu, J., Yao, Z., Sun, L., Shen, Y., Jin, Q., 2005. VFDB: a reference database for bacterial virulence factors. *Nucleic Acids Res.* 33 (suppl_1), D325–D328.
- Dijkshoorn, L., Nemec, A., Seifert, H., 2007. An increasing threat in hospitals: multidrug-resistant *Acinetobacter baumannii*. *Nat. Rev. Microbiol.* 5 (12), 939–951.
- Dolma, K.G., Khati, R., Paul, A.K., Rahmatullah, M., de Lourdes, Pereira M., Wilairatana, P., Khandelwal, B., Gupta, C., Gautam, D., Gupta, M., Goyal, R.K., 2022. Virulence characteristics and emerging therapies for biofilm-forming *Acinetobacter baumannii*: a review. *Biology* 11 (9), 1343.
- Fisher, E.L., Otto, M., Cheung, G.Y., 2018. Basis of virulence in enterotoxin-mediated staphylococcal food poisoning. *Front. Microbiol.* 9, 436.
- Gedefie, A., Demsis, W., Ashagrie, M., Kassa, Y., Tesfaye, M., Tilahun, M., Bisetegn, H., Sahle, Z., 2021. *Acinetobacter baumannii* biofilm formation and its role in disease pathogenesis: a review. *Infect. Drug Resist.* 3711–3719.
- Howard, A., O'Donoghue, M., Feeney, A., Sleator, R.D., 2012. *Acinetobacter baumannii*: an emerging opportunistic pathogen. *Virulence* 3 (3), 243–250.
- Hyatt, D., Chen, G.L., Locascio, P.F., Land, M.L., Larimer, F.W., Hauser, L.J., 2010. Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC Bioinform.* 11, 1.
- Imperi, F., Antunes, L.C., Blom, J., Villa, L., Iacono, M., Visca, P., Carattoli, A., 2011. The genomics of *Acinetobacter baumannii*: insights into genome plasticity, antimicrobial resistance and pathogenicity. *IUBMB Life* 63 (12), 1068–1074.
- Javkar, K., Rand, H., Hoffmann, M., Luo, Y., Sarria, S., Thirunavukkarasu, N., Pillai, C.A., McGann, P., Johnson, J.K., Strain, E., Pop, M., 2021. Whole-genome assessment of clinical *Acinetobacter baumannii* isolates uncovers potentially novel factors influencing carbapenem resistance. *Front. Microbiol.* (12), 714284.
- Jolley, K.A., Bray, J.E., Maiden, M.C., 2018. Open-access bacterial population genomics: BIGSdb software, the PubMLST. Org website and their applications. *Wellcome Open Res.* 3.
- Karah, N., Sundsfjord, A., Towner, K., Samuelsen, Ø., 2012. Insights into the global molecular epidemiology of carbapenem non-susceptible clones of *Acinetobacter baumannii*. *Drug Resist. Updat.* 15 (4), 237–247.
- Lee, C.R., Lee, J.H., Park, M., Park, K.S., Bae, I.K., Kim, Y.B., Cha, C.J., Jeong, B.C., Lee, S.H., 2017. Biology of *Acinetobacter baumannii*: pathogenesis, antibiotic resistance mechanisms, and prospective treatment options. *Front. Cell. Infect. Microbiol.* 7, 55.
- Liu, H.Y., Liu, H.X., Zheng, J.S., Cai, D., 2022. Metabolic consequences of malnutrition: how to balance nutrients and genes. *Front. Nutr.* 9, 1028502.
- Maiden, M.C., Van Rensburg, M.J., Bray, J.E., Earle, S.G., Ford, S.A., Jolley, K.A., McCarthy, N.D., 2013. MLST revisited: the gene-by-gene approach to bacterial genomics. *Nat. Rev. Microbiol.* 11 (10), 728–736.
- McArthur, A.G., Waglechner, N., Nizam, F., Yan, A., Azad, M.A., Baylay, A.J., Bhullar, K., Canova, M.J., De Pascale, G., Ejim, L., Kalan, L., 2013. The comprehensive antibiotic resistance database. *Antimicrob. Agents Chemother.* 57 (7), 3348–3357.
- Moubareck, C.A., Halat, D.H., 2020. Insights into *Acinetobacter baumannii*: A Review of Microbiological, Virulence, and Resistance Traits in a.
- Mortensen, B.L., Skaar, E.P., 2013. The contribution of nutrient metal acquisition and metabolism to *Acinetobacter baumannii* survival within the host. *Front. Cell. Infect. Microbiol.* (3), 95.
- Nocera, F.P., Attili, A.R., De Martino, L., 2021. *Acinetobacter baumannii*: its clinical significance in human and veterinary medicine. *Pathogens* 10 (2), 127.
- Olson, R.D., Assaf, R., Brettin, T., Conrad, N., Cucinell, C., Davis, J.J., Dempsey, D.M., Dickerman, A., Dietrich, E.M., Kenyon, R.W., Kuscoglu, M., 2023. Introducing the bacterial and viral bioinformatics resource center (BV-BRC): a resource combining PATRIC, IRD and ViPR. *Nucleic Acids Res.* 51 (D1), D678–D689.
- Page, A.J., Cummins, C.A., Hunt, M., Wong, V.K., Reuter, S., Holden, M.T., Fookes, M., Falush, D., Keane, J.A., Parkhill, J., 2015. Roary: rapid large-scale prokaryote pan genome analysis. *Bioinformatics* 31 (22), 3691–3693.
- Sahl, J.W., Del Franco, M., Pournaras, S., Colman, R.E., Karah, N., Dijkshoorn, L., Zarrilli, R., 2015. Phylogenetic and genomic diversity in isolates from the globally distributed *Acinetobacter baumannii* ST25 lineage. *Sci. Rep.* 5 (1), 15188.
- Sarshar, M., Behzadi, P., Scribano, D., Palamara, A.T., Ambrosi, C., 2021. *Acinetobacter baumannii*: an ancient commensal with weapons of a pathogen. *Pathogens* 10 (4), 387.
- Schmiel, D.H., Miller, V.L., 1999. Bacterial phospholipases and pathogenesis. *Microbes Infect.* 1 (13), 1103–1112.
- Seemann, T., 2014. Prokka: rapid prokaryotic genome annotation. *Bioinformatics* 30 (14), 2068–2069.
- Shamsizadeh, Z., Nikaeen, M., Nasr Esfahani, B., Mirhoseini, S.H., Hatamzadeh, M., Hassanzadeh, A., 2017. Detection of antibiotic resistant *Acinetobacter baumannii* in various hospital environments: potential sources for transmission of *Acinetobacter* infections. *Environ. Health Prev. Med.* 22, 1–7.
- Sharma, A.K., Dhasmana, N., Dubey, N., Kumar, N., Gangwal, A., Gupta, M., Singh, Y., 2017. Bacterial virulence factors: secreted for survival. *Indian J. Microbiol.* 57, 1.
- Sunshine, R.H., Wright, M.O., Maragakis, L.L., Harris, A.D., Song, X., Hebdon, J., Cosgrove, S.E., Anderson, A., Carnell, J., Jernigan, D.B., Kleinbaum, D.G., 2007. Multidrug-resistant *Acinetobacter* infection mortality rate and length of hospitalization. *Emerg. Infect. Dis.* 13 (1), 97.
- Tamura, K., Stecher, G., Kumar, S., 2021. MEGA11: molecular evolutionary genetics analysis version 11. *Mol. Biol. Evol.* 38 (7), 3022–3027.

- Tettelin, H., Riley, D., Cattuto, C., Medini, D., 2008. Comparative genomics: the bacterial pan-genome. *Curr. Opin. Microbiol.* 11 (5), 472–477.
- Touchon, M., Cury, J., Yoon, E.J., Krizova, L., Cerqueira, G.C., Murphy, C., Feldgarden, M., Wortman, J., Clermont, D., Lambert, T., Grillot-Courvalin, C., 2014. The genomic diversification of the whole *Acinetobacter* genus: origins, mechanisms, and consequences. *Genome Biol. Evol.* 6 (10), 2866–2882.
- Urhan, A., Abeel, T., 2021. A comparative study of pan-genome methods for microbial organisms: *Acinetobacter baumannii* pan-genome reveals structural variation in antimicrobial resistance-carrying plasmids. *Microbial. Genomics.* 7 (11).
- Verma, P., Tiwari, M., Tiwari, V., 2021. Efflux pumps in multidrug-resistant *Acinetobacter baumannii*: current status and challenges in the discovery of efflux pumps inhibitors. *Microb. Pathog.* (152), 104766.
- Willenbrock, H., Hallin, P.F., Wassenaar, T.M., Ussery, D.W., 2007. Characterization of probiotic *Escherichia coli* isolates with a novel pan-genome microarray. *Genome Biol.* 8 (12), 1–6.
- Woodford, N., Turton, J.F., Livermore, D.M., 2011. Multiresistant gram-negative bacteria: the role of high-risk clones in the dissemination of antibiotic resistance. *FEMS Microbiol. Rev.* 35 (5), 736–755.
- Wu, X., Wan, X.F., Wu, G., Xu, D., Lin, G., 2006. Phylogenetic analysis using complete signature information of whole genomes and clustered neighbour-joining method. *Int. J. Bioinform. Res. Appl.* 2 (3), 219–248.